Vector Spaces III: Sums, Intersections and Direct Sums: Answer Key

Problem sheet for the Day 4 class · CSIR NET / GATE / IIT JAM previous year questions

Answers

Q	1	2	3	4	5	6	7	8	9	10
Answer	3	60	(C)	(A)	(B),(C)	(B),(C)	(C),(D)	(A),(C),(D)	1	(A),(C)

Hints and short solutions

1. **3** GATE 2013 Two distinct planes through 0; their sum is $\mathbb{R}^3$ (and their intersection a line).
2. **60** JAM 2019 The intersection kills coordinates divisible by 4 or 5: $25 + 20 - 5 = 40$; $100 - 40 = 60$.
3. **(C)** CSIR NET Dec 2018 W_1: the third equation is $2\times$first $-$ second, so $\dim W_1 = 2$. W_2: $x = 0 = v$, $w = -u$, so $\dim W_2 = 1$, and $W_2 = \operatorname{span}\{(1, 0, -1, 0)\} \subset W_1$. Hence $\dim(W_1 \cap W_2) = 1$, $\dim(W_1 + W_2) = 2$.
4. **(A)** CSIR NET Dec 2018 (B), (D) do not contain $(0, 1, 1)$. In (C), $W \cap W_1 = \operatorname{span}\{(1, 0, -1)\}$, not orthogonal to $(0, 1, 1)$. In (A), $W \cap W_1 = \operatorname{span}\{(0, 1, -1)\} \perp (0, 1, 1)$.
5. **(B),(C)** CSIR NET Dec 2016 $\dim(W_1 \cap W_2) = 8$ (distinct hyperplanes), and intersecting with one more hyperplane drops the dimension by at most 1: $7 \le \dim W \le 8$.
6. **(B),(C)** CSIR NET Dec 2014 (B), (C) follow from the definitions. Three distinct lines U, V, W in $\mathbb{R}^2$ break (A) and (D): $U \cap (V + W) = U$ but $U \cap V + U \cap W = 0$.
7. **(C),(D)** Exam pattern $12 = 19 + 17 - 24 \le \dim(S \cap T) \le 17$; $19 \le \dim(S + T) \le 24$.
8. **(A),(C),(D)** Exam pattern (B): both contain the diagonal matrices, so the sum is not direct.
9. **1** Exam pattern $(1, 1, 1, 1)$ lies in both; the four vectors have rank 3, so $\dim(W_1 + W_2) = 3$ and $\dim(W_1 \cap W_2) = 2 + 2 - 3 = 1$.
10. **(A),(C)** Exam pattern A line is a complement of the hyperplane iff its direction is not in W, i.e. its coordinate sum is non-zero.